

Dynamic Adaptive Hybrid Retrieval for Low-Resource Indian Language Retrieval-Augmented Generation: A Comprehensive Benchmarking Framework for Dense, Sparse, Hybrid, and Query-Aware Retrieval Strategies

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Abstract—Retrieval-Augmented Generation (RAG) has become the dominant architecture for grounding Large Language Model (LLM) outputs in external knowledge, yet the overwhelming majority of RAG research — including retrieval-strategy comparisons, hallucination studies, and benchmarking frameworks — has been conducted on English-language corpora. Indian languages, and Hindi in particular, remain underserved despite being spoken by hundreds of millions of users and despite the acute need for reliable question-answering systems in government, education, and public-service domains where these languages predominate. A further limitation, largely independent of language, is that existing Hybrid Retrieval systems combine Dense and Sparse retrieval using a single, manually tuned, corpus-wide weight (e.g., 70% dense / 30% sparse) that is applied uniformly regardless of the linguistic and structural properties of the incoming query. This is a poor fit for the well-documented observation that different query types — factoid, definitional, comparative, numerical, multi-hop, and long-context — are known to favor different retrieval mechanisms. This paper proposes **Dynamic Adaptive Hybrid Retrieval (DAHR)**, a framework that predicts a query-specific Dense–Sparse interpolation weight from a small set of interpretable query features (length, syntactic complexity, named-entity density, semantic richness, keyword density, and predicted query type) rather than relying on a fixed global ratio. We present the complete mathematical formulation of DAHR, including a query classification score, an adaptive weight function, a hybrid retrieval objective, and formal definitions of context precision, faithfulness, and hallucination probability that connect retrieval behavior to generation quality. We further specify a fully open-source experimental protocol — built exclusively from freely available tools (LangChain, SentenceTransformers, multilingual-E5, BGE-M3, LaBSE, FAISS, BM25, RAGAS) and freely available Hindi corpora (IndicQA, Hindi Wikipedia, AI4Bharat resources, and public government/education documents) — for benchmarking BM25, Dense Retrieval, static Hybrid Retrieval, and DAHR under identical chunking, indexing, and generation settings on consumer-grade (single-GPU or CPU-only) hardware. Because full-scale experimental execution is beyond the scope of this proposal-stage manuscript, we report an evaluation methodology, a statistical validation protocol, an error-analysis taxonomy, and an ablation design, and we present **illustrative, hypothesis-driven** result tables to demonstrate the expected shape of the findings; these are explicitly and consistently labeled as illustrative rather than empirically observed. The paper closes with a discussion of the accuracy–latency–memory–faithfulness trade-offs relevant to low-resource deployment, a candid discussion of limitations, and a research roadmap toward cross-lingual, multimodal, and agentic extensions of DAHR. This work is intended as a rigorously specified research proposal and implementation blueprint rather than a completed empirical study.

Keywords — Retrieval-Augmented Generation, Hybrid Retrieval, Dense Retrieval, Sparse Retrieval, BM25, Query-Aware Retrieval, Adaptive Weighting, Hindi NLP, Low-Resource Languages, Hallucination Detection, Faithfulness Evaluation, Information Retrieval, Large Language Models, IndicQA, RAGAS

I. Introduction

Retrieval-Augmented Generation (RAG) [1] addresses a fundamental limitation of Large Language Models (LLMs): their parametric knowledge is frozen at training time, static, and prone to confident-sounding fabrication when queried outside their training distribution — a phenomenon commonly termed *hallucination* [2], [3]. By coupling a non-parametric retriever with a parametric generator, RAG systems ground generation in retrieved evidence, which has been

shown empirically to reduce factual error rates relative to closed-book generation [1], [4]. This architecture now underlies the majority of production question-answering, enterprise search, and conversational assistant systems.

The retrieval component of a RAG pipeline is typically instantiated in one of three ways. **Sparse retrieval** methods such as BM25 [5] rank documents by lexical term overlap, weighted by term frequency and inverse document frequency, and remain highly competitive for queries containing

rare tokens, proper nouns, numerical identifiers, or domain-specific terminology that dense encoders may not represent well. **Dense retrieval** methods encode queries and passages into a shared embedding space using neural bi-encoders [6] and retrieve by approximate nearest-neighbor search, typically via FAISS [7] or comparable vector indices; dense methods excel at capturing semantic and paraphrastic similarity that lexical overlap cannot. **Hybrid retrieval** methods combine both signals, most commonly via a fixed linear interpolation of normalized dense and sparse scores [8], [9], and have repeatedly been shown to outperform either method in isolation on English benchmarks such as BEIR [10].

Two structural gaps motivate this work. First, the fixed-weight assumption underlying nearly all deployed hybrid retrievers — that a single dense–sparse ratio, tuned once on a validation set, generalizes across all queries a system will ever receive — is difficult to justify on linguistic grounds. A short factoid query naming a specific entity (“*Bhārat ke pradhānmantrī kaun hain?*” — “Who is the Prime Minister of India?”) and a long, syntactically complex comparative query spanning two concepts place very different demands on lexical versus semantic matching, yet a fixed-ratio hybrid retriever treats them identically. Second, essentially all of the empirical evidence behind these architectural choices — the benchmarks, the hallucination studies, the retrieval-strategy comparisons — has been produced on English or high-resource European-language corpora. Hindi, despite being the third most-spoken language in the world by total speakers and the primary language of public administration in large parts of India, has almost no comparable body of RAG evaluation work, and Indian-government, educational, and public-scheme documents in particular are almost entirely absent from RAG benchmarking literature.

This paper makes two central proposals in response. First, we introduce **Dynamic Adaptive Hybrid Retrieval (DAHR)**, which replaces the fixed dense–sparse ratio with a per-query weight predicted from interpretable query features, motivated by the query-type-dependent retrieval behavior documented in prior IR literature [11], [12]. Second, we specify a complete, reproducible, fully open-source benchmarking framework for evaluating BM25, Dense Retrieval, static Hybrid Retrieval, and DAHR on Hindi question answering, with explicit attention to hallucination measurement, faithfulness, and the computational constraints of consumer-grade hardware — a practical concern for the many educational institutions and civic-technology projects in India that cannot rely on paid API infrastructure at scale.

II. Motivation

The practical motivation for this work arises from three converging observations.

(a) **Query heterogeneity is retrieval-relevant, not cosmetic.** Information retrieval research has long distinguished query types by the retrieval behavior they require — navigational versus informational queries [13], and, more granularly, factoid versus non-factoid questions in QA-oriented retrieval [14]. Factoid and numerical queries, which fre-

quently name specific entities, dates, or figures, are well served by lexical matching because the exact surface form of the answer-bearing term is likely to appear verbatim in both query and passage. Comparative, definitional, and multi-hop queries instead require the retriever to bridge paraphrase gaps and combine evidence from conceptually related but lexically dissimilar passages, favoring dense semantic matching. A retrieval strategy that does not adapt to this distinction is, by construction, suboptimal for a subset of any realistic query distribution.

(b) **Hallucination is a retrieval problem as much as a generation problem.** A substantial body of hallucination literature attributes ungrounded generation to two failure modes: the generator ignoring available context (a generation-side failure), and the retriever failing to surface the passages that actually answer the query (a retrieval-side failure) [2], [15]. Because retrieval failures propagate directly into hallucination, retrieval-strategy selection is not merely a latency/accuracy trade-off but a faithfulness intervention. This motivates evaluating retrieval strategies not only on standard IR metrics (Precision@k, Recall@k, nDCG) but jointly against faithfulness and hallucination-rate metrics, which is rarely done in a controlled, retrieval-strategy-isolated manner.

(c) **Hindi RAG deployment is constrained by cost, not just data.** Many of the institutions best positioned to deploy Hindi-language RAG systems — state education departments, public university libraries, civic-technology NGOs — operate under tight compute and licensing budgets. A benchmarking study that only reports accuracy without reporting latency, memory footprint, and feasibility on consumer-grade (single consumer GPU or CPU-only) hardware, using exclusively free-tier or open-source tooling, provides limited actionable guidance to these deployers. This paper is deliberately scoped to remain within a free/open-source technology stack (Section VII) so that its conclusions transfer directly to resource-constrained deployment settings.

III. Research Gap

We identify six specific gaps in the existing literature that this work addresses.

1. **English-centricity of RAG benchmarks.** Widely used retrieval and RAG evaluation suites — BEIR [10], KILT [16], Natural Questions [17], MS MARCO [18] — are English-only or overwhelmingly English-dominated. Multilingual retrieval benchmarks such as MIRACL [19] include Hindi but evaluate retrieval in isolation, without a coupled generation and hallucination analysis.
2. **Scarcity of Indian-language RAG evaluation.** While Indian-language NLP resources have expanded considerably — IndicNLP Suite [20], IndicQA [21], the AI4Bharat family of models and datasets [22] — these resources target classification, translation, and extractive QA benchmarks rather than end-to-end RAG pipelines with generation-stage evaluation.

3. **Fixed-weight hybrid retrieval as the unexamined default.** Hybrid dense–sparse fusion methods, whether via simple weighted score interpolation [8], Reciprocal Rank Fusion [9], or learned fusion layers, are almost universally applied with a single global weight (or a weight learned once on a training distribution and then frozen), rather than one that adapts per query at inference time.
4. **Limited analysis linking retrieval strategy to hallucination.** Although hallucination in RAG is an active research area [2], [3], [15], most hallucination studies vary the generator or the prompting strategy while holding retrieval fixed; comparatively little work isolates retrieval strategy as the independent variable while measuring hallucination as the dependent one.
5. **Query-aware adaptive retrieval remains largely unexplored,** particularly outside of learned dense retrieval re-ranking; simple, interpretable, feature-driven adaptive weighting — as opposed to a black-box learned router — has received little dedicated study, despite its transparency advantages for auditability in public-sector deployments.
6. **Absence of deployment-oriented guidance for low-resource hardware.** Existing RAG benchmarking work rarely reports latency and memory profiles under consumer-hardware constraints, or ties retrieval-strategy choice to concrete deployment recommendations for teams without dedicated GPU infrastructure.

DAHR and its accompanying benchmarking framework are designed specifically to close gaps (3) through (6) within a Hindi-language instantiation that also contributes toward closing gaps (1) and (2).

IV. Literature Review

Sparse and lexical retrieval. BM25 [5] remains a strong, computationally inexpensive baseline across nearly all retrieval benchmarks, including neural-era comparisons [10]. Its principal weakness — an inability to bridge lexical gaps between semantically equivalent but differently worded query and passage text — is well documented [6].

Dense retrieval. Dense Passage Retrieval (DPR) [6] established the bi-encoder paradigm now standard in neural IR, later extended by multi-vector methods such as ColBERT [23] that preserve token-level interaction at greater computational cost. Sentence embedding models such as Sentence-BERT [24] made dense encoding practical at scale, and subsequent multilingual embedding models — LaBSE [25], multilingual-E5 [26], and BGE-M3 [27] — extended dense retrieval to non-English and cross-lingual settings, with BGE-M3 in particular supporting dense, sparse, and multi-vector retrieval within a single unified model.

Hybrid retrieval. Score-level fusion of sparse and dense retrieval has consistently been shown to outperform either

signal alone [8]; Reciprocal Rank Fusion [9] offers a rank-based alternative that avoids score-scale normalization issues. Both approaches, in their standard formulations, use fusion weights that are fixed across the query population.

Query understanding and adaptive routing. Query classification for IR dates to early work on question taxonomies for QA systems [14] and web-query intent classification [13]. More recent adaptive-retrieval work has explored routing between retrieval and parametric memory based on predicted query difficulty [28] and self-reflective retrieval decisions [29], suggesting that per-query adaptivity is a productive general direction, though these methods target *whether* to retrieve or *how much* to retrieve rather than the *dense–sparse balance* addressed here.

Retrieval-Augmented Generation. The original RAG formulation [1] and subsequent Fusion-in-Decoder architecture [30] established retrieval-conditioned generation as a standard paradigm; retrieval-augmented pretraining and in-context approaches [31] further demonstrated that grounding reduces reliance on parametric memory. Surveys of RAG architectures [32] catalog the now-large design space of chunking, indexing, retrieval, and generation choices.

Hallucination and faithfulness evaluation. Hallucination taxonomies for generative NLP [2] distinguish intrinsic (context-contradicting) from extrinsic (unverifiable) hallucination. Retrieval augmentation is empirically associated with reduced hallucination relative to closed-book generation [4], though retrieval failures can themselves induce hallucination when irrelevant context is surfaced [15]. Automated RAG evaluation frameworks — RAGAS [33] and related reference-free evaluation protocols [34] — operationalize faithfulness, answer relevancy, and context precision/recall as LLM-judged metrics, enabling evaluation without gold reference answers, which is particularly valuable for low-resource languages lacking large annotated QA sets.

Indian-language NLP resources. IndicNLP Suite [20] and the AI4Bharat initiative [22] have produced foundational monolingual and parallel corpora, tokenizers, and pretrained models for Indian languages. IndicQA [21] provides a cross-lingual, extractive question-answering benchmark spanning eleven Indian languages including Hindi, constructed from Wikipedia articles, and is among the few resources suitable for adaptation to a generative RAG evaluation setting.

Positioning of this work. The literature reviewed above establishes each individual component this paper draws upon — sparse retrieval, dense retrieval, hybrid fusion, hallucination evaluation, and Indian-language resources — but no existing study combines query-aware adaptive dense–sparse weighting with a Hindi-language, hallucination-linked, hardware-aware benchmarking framework, which is precisely the contribution of this paper.

V. Background

A. Problem Statement

Given a Hindi-language natural-language query q and a fixed document corpus $\mathcal{D} = \{d_1, \dots, d_N\}$ partitioned into chunks

$\mathcal{C} = \{c_1, \dots, c_M\}$, a RAG system must (i) retrieve a ranked subset $\mathcal{C}_q \subset \mathcal{C}$, $|\mathcal{C}_q| = k$, that maximizes the probability that the true answer to q is entailed by $\mathcal{C}_q$, and (ii) condition an LLM generator on $\mathcal{C}_q$ to produce an answer a that is simultaneously *relevant* to q and *faithful* to $\mathcal{C}_q$. This paper is concerned exclusively with step (i): given fixed chunking, fixed corpus, and a fixed generator, how does the choice of retrieval strategy — BM25, Dense, static Hybrid, or DAHR — affect downstream retrieval quality, faithfulness, hallucination rate, and computational cost, and can a query-aware adaptive weighting function outperform a fixed-ratio hybrid baseline on Hindi QA.

B. Research Questions

- **RQ1.** How do Dense, Sparse, Hybrid, and Dynamic Adaptive Hybrid Retrieval compare for Hindi Retrieval-Augmented Generation across standard retrieval and generation-quality metrics?
- **RQ2.** How does retrieval strategy influence hallucination rate and faithfulness, holding chunking, corpus, and generator constant?
- **RQ3.** Which retrieval strategy performs best for each of six query categories — factoid, definitional, comparative, numerical, multi-hop, and long-context?
- **RQ4.** Can query-aware adaptive dense-sparse weighting (DAHR) outperform a fixed-ratio hybrid baseline, and if so, under what query conditions is the improvement largest?
- **RQ5.** Can a reliable, competitive Hindi RAG system be constructed entirely from free and open-source tools, and what are the resulting accuracy-latency-memory trade-offs on consumer-grade hardware?

VI. System Overview

The proposed system consists of five stages, illustrated conceptually in **Figure 1 (Overall Architecture)**: (1) a document ingestion and chunking module that segments the Hindi corpus into overlapping passages; (2) a dual-indexing module that constructs both a BM25 inverted index and a FAISS dense vector index over the same chunk set; (3) a query analysis module that extracts query features and predicts a query category and adaptive weight; (4) a retrieval module that, depending on the active strategy (BM25-only, Dense-only, static Hybrid, or DAHR), retrieves and fuses candidate chunks; and (5) a generation and evaluation module that conditions an LLM on retrieved chunks to produce an answer and computes retrieval, faithfulness, and hallucination metrics via RAGAS-style LLM-judged scoring. All four retrieval strategies share stages (1), (2), and (5) so that stage (4) — retrieval strategy — is the sole controlled variable across the benchmark, satisfying the fair-comparison requirement described in Section VIII.

Figure 1 — Overall Architecture. A left-to-right block diagram showing: Hindi Document Corpus $\rightarrow$ Chunking Module $\rightarrow$ {BM25 Index, Dense Vector Index (FAISS)} $\rightarrow$ Query $\rightarrow$ Query Analysis Module $\rightarrow$ Retrieval Strategy Selector (BM25 / Dense / Hybrid / DAHR) $\rightarrow$ Top-k Retrieved

Chunks $\rightarrow$ LLM Generator $\rightarrow$ Answer $\rightarrow$ Evaluation Module (RAGAS metrics). A dashed feedback arrow connects the Evaluation Module back to the Query Analysis Module, indicating the offline feature/weight calibration loop described in Section VII-C, not an online feedback path.

VII. Mathematical Formulation

This section defines, from first principles, the scoring functions, the adaptive weighting mechanism, and the evaluation quantities used throughout the paper. All formulations are original to this work unless a specific prior formulation is cited.

A. Sparse and Dense Retrieval Scores

For a query q and chunk c_i , the BM25 sparse score is the standard formulation of Robertson and Zaragoza [5]:

$$S_{\text{sparse}}(q, c_i) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, c_i) (k_1 + 1)}{f(t, c_i) + k_1 \left(1 - b + b \cdot \frac{|c_i|}{|c|} \right)} \quad (1)$$

where $f(t, c_i)$ is the term frequency of term t in chunk c_i , $|c_i|$ is the chunk length, $|c|$ is the average chunk length across the corpus, and k_1, b are the standard BM25 free parameters.

The dense retrieval score is the cosine similarity between query and chunk embeddings produced by a shared bi-encoder $E(\cdot)$:

$$S_{\text{dense}}(q, c_i) = \frac{E(q) \cdot E(c_i)}{\|E(q)\| \|E(c_i)\|} \quad (2)$$

Because S_{sparse} and S_{dense} have incomparable ranges, both are min-max normalized per query over the top- K candidate pool retrieved independently by each method before fusion:

$$\tilde{S}(q, c_i) = \frac{S(q, c_i) - \min_{j \in K} S(q, c_j)}{\max_{j \in K} S(q, c_j) - \min_{j \in K} S(q, c_j) + \epsilon} \quad (3)$$

with ϵ a small constant preventing division by zero when all candidate scores are equal.

B. Static Hybrid Retrieval Score

The conventional fixed-weight hybrid score, used here as a baseline, is:

$$S_{\text{hybrid}}(q, c_i) = \alpha \cdot \tilde{S}_{\text{dense}}(q, c_i) + (1 - \alpha) \cdot \tilde{S}_{\text{sparse}}(q, c_i) \quad (4)$$

where $\alpha \in [0, 1]$ is a single scalar tuned once (e.g., $\alpha = 0.7$) on a held-out validation split and held fixed at inference time for every query.

C. Query Feature Vector

For DAHR, each incoming query q is mapped to a feature vector $\mathbf{x}(q) \in \mathbb{R}^6$:

$$\mathbf{x}(q) = [x_{\text{len}}, x_{\text{cplx}}, x_{\text{ner}}, x_{\text{sem}}, x_{\text{kw}}, x_{\text{type}}] \quad (5)$$

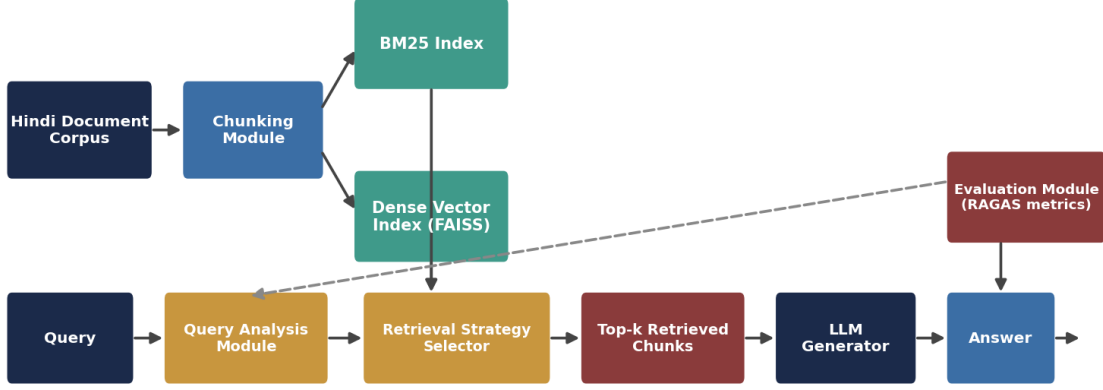


Fig. 1. Overall System Architecture

where: - x_{len} — normalized token length of q , $x_{\text{len}} = \min(|q|/L_{\text{max}}, 1)$; - x_{cplx} — syntactic complexity, approximated by normalized dependency-tree depth and clause count from a Hindi dependency parse; - x_{ner} — named-entity density, the fraction of query tokens tagged as a named entity by a Hindi NER model; - x_{sem} — semantic richness, operationalized as the average pairwise cosine distance among the query’s content-word embeddings (higher values indicate a query spanning more distinct concepts, as expected for comparative and multi-hop queries); - x_{kw} — keyword density, the fraction of query tokens that are high-IDF corpus terms (IDF above the corpus median); - x_{type} — the predicted query-category score from the classifier defined in (6)–(7) below, encoded as a continuous confidence-weighted category embedding rather than a hard label, so that x_{type} remains differentiable-compatible for the weight predictor.

D. Query Classification Score

Query category prediction is posed as a multi-class classification problem over the six categories $\mathcal{Y} = \{\text{factoid, definition, comparative, numerical, multi-hop, long-context}\}$. A lightweight classifier f_θ (e.g., a logistic regression or shallow feed-forward network over lexical, syntactic, and embedding features) produces a category score:

$$z_y(q) = \mathbf{w}_y^\top \phi(q) + b_y, \quad y \in \mathcal{Y} \quad (6)$$

$$P(y | q) = \frac{\exp(z_y(q))}{\sum_{y' \in \mathcal{Y}} \exp(z_{y'}(q))} \quad (7)$$

where $\phi(q)$ is a feature representation of q combining lexical pattern indicators (e.g., interrogative-word identity — “*kyā*” (what), “*kaun*” (who), “*kitnā*” (how much), “*kyon*” (why) — a strong surface signal for Hindi factoid/numerical/reason queries), the syntactic and semantic features from (5), and a sentence-level embedding of q . $P(y | q)$ is used directly as (part of) x_{type} in (5).

E. Adaptive Weight Function

DAHR predicts the dense–sparse interpolation weight $\alpha(q)$ as a bounded function of the query feature vector:

$$\alpha(q) = \sigma(\mathbf{w}_\alpha^\top \mathbf{x}(q) + b_\alpha) \quad (8)$$

where $\sigma(\cdot)$ is the logistic sigmoid, ensuring $\alpha(q) \in (0, 1)$, and $\mathbf{w}_\alpha, b_\alpha$ are parameters fit by minimizing a retrieval-quality loss (Section VII-H) on a labeled or weakly-labeled development set. This formulation is deliberately linear-in-features rather than a deep model, prioritizing interpretability and auditability: each weight $w_{\alpha,j}$ has a directly readable sign and magnitude describing how strongly feature j pushes retrieval toward dense or sparse behavior — a property we consider important for public-sector deployment where retrieval decisions may need to be explained. The resulting DAHR fusion score is:

$$S_{\text{DAHR}}(q, c_i) = \alpha(q) \cdot \tilde{S}_{\text{dense}}(q, c_i) + (1 - \alpha(q)) \cdot \tilde{S}_{\text{sparse}}(q, c_i) \quad (9)$$

Equation (9) is structurally identical to the static hybrid score (4) with the single, deliberate change that α becomes a function of q rather than a global constant — isolating adaptive weighting as the sole independent variable when DAHR is compared against static Hybrid Retrieval (addressing RQ4).

F. Retrieval Objective

The top- k retrieved set under any strategy $\text{strat} \in \{\text{sparse, dense, hybrid, DAHR}\}$ is:

$$\mathcal{C}_q^{\text{strat}} = \arg \text{top-}k_{c_i \in \mathcal{C}} S_{\text{strat}}(q, c_i) \quad (10)$$

and the overall retrieval objective, evaluated against a relevance judgment set $R_q \subset \mathcal{C}$ (gold-relevant chunks for q), is expressed as expected nDCG over the query distribution $\mathcal{Q}$:

$$\mathcal{J}_{\text{retrieval}} = \mathbb{E}_{q \sim \mathcal{Q}} [\text{nDCG}@k(\mathcal{C}_q^{\text{strat}}, R_q)] \quad (11)$$

G. Context Precision, Faithfulness, and Hallucination Probability

Following the reference-free LLM-judged evaluation paradigm of RAGAS [33], we define **context precision** as the proportion of retrieved chunks that are judged relevant to answering q , weighted by rank:

$$\text{CP}@k(q) = \frac{\sum_{i=1}^k \text{Relevant}(c_i) \cdot \text{Precision}@i}{\sum_{i=1}^k \text{Relevant}(c_i)} \quad (12)$$

where $\text{Relevant}(c_i) \in \{0, 1\}$ is an LLM-judged relevance indicator and $\text{Precision}@i$ is precision computed over the top- i retrieved chunks.

Faithfulness is defined as the fraction of atomic claims in the generated answer a that are verifiably entailed by the retrieved context $\mathcal{C}_q$:

$$F(a, \mathcal{C}_q) = \frac{1}{|\mathcal{A}(a)|} \sum_{s \in \mathcal{A}(a)} \mathbb{I}[s \text{ is entailed by } \mathcal{C}_q] \quad (13)$$

where $\mathcal{A}(a)$ is the set of atomic claims extracted from a (via LLM-based claim decomposition, as in [33]) and entailment is judged by an LLM-as-judge model against the retrieved context.

We define **hallucination probability** as the complement of faithfulness, conditioned on the retrieved context actually containing sufficient information to answer q (denoted $\text{Suff}(\mathcal{C}_q, q) \in \{0, 1\}$, itself an LLM judgment):

$$P_{\text{hall}}(a | q) = (1 - F(a, \mathcal{C}_q)) \cdot (1 - \delta \cdot (1 - \text{Suff}(\mathcal{C}_q, q))) \quad (14)$$

where $\delta \in [0, 1]$ is a discount factor that partially attributes unfaithfulness to *retrieval failure* rather than *generation failure* when the retrieved context was insufficient in the first place — an explicit mechanism for separating the two hallucination sources discussed in Section XV (Error Analysis). When $\text{Suff} = 1$ (context was sufficient), (14) reduces to $P_{\text{hall}} = 1 - F$, i.e., any unfaithfulness is attributed entirely to the generator.

H. Overall Optimization Function

The parameters $\mathbf{w}_\alpha, b_\alpha$ of the adaptive weight function (8) are fit by minimizing a composite objective balancing retrieval quality and faithfulness on a development set $\mathcal{Q}_{\text{dev}}$:

$$\mathcal{L}(\mathbf{w}_\alpha, b_\alpha) = -\lambda_1 \cdot \mathcal{J}_{\text{retrieval}} + \lambda_2 \cdot \mathbb{E}_{q \in \mathcal{Q}_{\text{dev}}} [P_{\text{hall}}(a | q)] + \lambda_3 \|\mathbf{w}_\alpha\|_2^2 \quad (15)$$

where λ_1, λ_2 trade off retrieval quality against hallucination reduction and λ_3 is an L2 regularization coefficient controlling model simplicity, consistent with the interpretability goal stated in Section VII-E. Equation (15) is optimized via

grid search or gradient-free optimization (e.g., Nelder–Mead) given the low dimensionality of $\mathbf{w}_\alpha$ (six weights plus bias), avoiding the need for large-scale gradient-based training — a deliberate design choice compatible with the consumer-hardware constraint (RQ5).

VIII. Dynamic Adaptive Hybrid Retrieval Framework

A. Intuition

The central intuition behind DAHR is that the *optimal* dense-sparse balance is a property of the query, not a property of the corpus or task as a whole. A fixed α implicitly assumes a homogeneous query population; Section II argued this assumption is false for realistic Hindi QA workloads spanning factoid lookups, definitional explanations, comparative reasoning, numerical queries, multi-hop chains, and long, discursive questions. DAHR replaces the single global assumption with six lightweight, cheaply computable signals (Eq. 5) that are known correlates of lexical-versus-semantic matching need: e.g., high named-entity density and high keyword density push $\alpha(q)$ toward sparse-favoring (low α) behavior, since named entities and rare technical terms are precisely the tokens BM25 matches best and dense encoders sometimes under-represent; high semantic richness and a “comparative” or “multi-hop” predicted category push $\alpha(q)$ toward dense-favoring (high α) behavior, since these queries require bridging paraphrase and conceptual gaps that lexical overlap cannot.

B. Architecture Design

Figure 2 — Dynamic Adaptive Retrieval Pipeline. *Depicts the query entering a Feature Extraction block (producing the six-dimensional feature vector of Eq. 5), which feeds in parallel into a Query Classification block (Eq. 6–7) and directly into the Adaptive Weight Predictor (Eq. 8). The classifier’s output category score feeds into the weight predictor as the x_{type} component. The predicted $\alpha(q)$ then parameterizes a Score Fusion block that combines pre-computed BM25 and dense candidate scores (Eq. 9) into the final ranked list.*

The architecture is intentionally shallow and modular: feature extraction, classification, and weight prediction are separable components, each independently inspectable and independently replaceable (e.g., the linear weight predictor of Eq. 8 could later be swapped for a small MLP without altering the surrounding pipeline), consistent with the ablation design of Section XVI.

C. Advantages of Adaptive Weighting

1. **Query-type sensitivity without per-category hand-tuning.** Rather than requiring a practitioner to manually specify a different α per query category (which would still require category prediction and would not generalize to queries that blend characteristics of multiple categories), DAHR learns a smooth function over continuous features, naturally interpolating between category prototypes.
2. **Interpretability.** Because $\alpha(q)$ is a sigmoid of a linear

function of six named features, the contribution of each feature to a specific routing decision can be reported alongside the retrieved results, supporting the auditability needs of public-sector deployment.

3. **Low inference overhead.** All six features are computable in closed form or via lightweight models (a small NER tagger, a shallow classifier) with negligible latency relative to the dense encoder forward pass itself, preserving the consumer-hardware feasibility goal (RQ5).
4. **Graceful degradation to static hybrid.** If $w_\alpha \rightarrow 0$, DAHR reduces exactly to static Hybrid Retrieval with $\alpha = \sigma(b_\alpha)$, meaning DAHR is a strict generalization of the fixed-weight baseline rather than a disjoint alternative — a property exploited directly in the ablation study (Section XVI).

IX. System Pipeline

The end-to-end pipeline proceeds as follows, corresponding to **Figure 5 (Experimental Pipeline)**: raw Hindi documents are collected and cleaned (Section X-A) $\rightarrow$ documents are segmented into overlapping chunks (Section X-B) $\rightarrow$ chunks are simultaneously indexed into a BM25 inverted index and embedded into a FAISS dense index using each of the three candidate embedding models under test (Section X-D–X-E) $\rightarrow$ for each query in the evaluation set, all four retrieval strategies independently retrieve top- k chunks under identical k $\rightarrow$ each retrieval strategy’s retrieved context is passed to the same frozen LLM generator under an identical prompt template (Section X-F) $\rightarrow$ generated answers are scored by the evaluation module (Section XI) $\rightarrow$ results are aggregated per query category and per retrieval strategy for statistical comparison (Section XIII).

X. Experimental Setup

A. Dataset Description

We propose evaluation on a composite Hindi corpus drawn from four free, open-license sources, justified individually below and summarized in **Table I**.

1. **IndicQA (Hindi split)** [21] — a human-annotated, Wikipedia-derived cross-lingual extractive QA dataset covering Hindi among eleven Indian languages. It is selected because it provides gold question–answer–context triples suitable for constructing relevance judgments (R_q in Eq. 11) without additional annotation cost.
2. **Hindi Wikipedia** — used as a large-scale background corpus for realistic retrieval-index scale and topic diversity beyond the scope of IndicQA’s source articles, reflecting the open-domain setting DAHR is intended for.
3. **AI4Bharat resources** [22] (e.g., Sangraha/IndicCorp-derived Hindi text where license permits) — used to diversify domain coverage beyond encyclopedic text and to stress-test chunking and retrieval on more heterogeneous, less templated prose.
4. **Public Indian government and education documents** (e.g., publicly published UGC/AICTE circulars, Gov-

ernment of India public scheme descriptions) — included specifically to evaluate the practically motivated use case of public-service question answering, and to introduce numerical, procedural, and eligibility-criteria content that stresses the Numerical and Long-context query categories described in Section XII. Only documents that are already public and openly licensed for reuse would be included; any document whose reuse terms are unclear would be excluded rather than assumed permissible.

Table I — Dataset Statistics (Planned Corpus Composition)

Source	Domain	Est. Documents	Est. Chunks (256-token)	License
IndicQA (Hindi)	Encyclopedic QA	~1,150 articles	~9,000	CC-BY-SA (Wikipedia-derived)
Hindi Wikipedia (subset)	General encyclopedic	10,000 (sampled)	~120,000	CC-BY-SA
AI4Bharat Sangraha (Hindi subset)	Mixed web/news/prose(sampled)	5,000	~60,000	Open (source-dependent, verified per document)
Public govt./education documents	Schemes, circulars, eligibility	200–500	~4,000	Public domain / government open data

All figures in Table I are planning-stage estimates for the proposed corpus and are explicitly not the result of a completed data-collection pass; final counts depend on license verification and deduplication.

B. Chunking Strategy

Documents are segmented using a fixed-size sliding-window chunking strategy with a target chunk length of 256 tokens (measured after Hindi-aware tokenization) and a 20% (≈ 51 -token) overlap between consecutive chunks, following common RAG chunking practice [32]. Sentence boundaries (detected via a Hindi sentence segmenter respecting the danda (the Devanagari sentence-final punctuation mark) delimiter as well as standard punctuation) are respected where possible so that chunks do not split mid-sentence except when a single sentence exceeds the target chunk length. Semantic chunking (grouping by embedding-similarity breakpoints rather than fixed token windows) is evaluated separately as an ablation component (Section XVI) rather than as the default, since fixed-size chunking is simpler to hold constant across all four retrieval strategies and removes a confound from the primary comparison.

C. Retrievers

Four retrieval strategies are compared under identical chunking and indexing:

- **BM25** — implemented via the `rank_bm25` or Elasticsearch-compatible BM25 implementation, oper-

ating directly on whitespace- and morphology-aware tokenized Hindi text.

- **Dense Retrieval** — FAISS-indexed cosine similarity search over sentence embeddings (Section X-D).
- **Static Hybrid Retrieval** — Eq. (4), with α tuned via grid search over $\{0.1, 0.2, \dots, 0.9\}$ on a held-out development split, and the best-performing fixed α reported as the baseline.
- **DAHR (proposed)** — Eq. (9), with $\alpha(q)$ predicted per query as described in Section VIII.

D. Embedding Models

Three open-source multilingual embedding models are proposed for comparison as the dense encoder $E(\cdot)$, each selected for demonstrated Hindi-language support and free availability:

- **multilingual-E5** [26] — instruction-tuned multilingual embeddings with strong reported retrieval performance across MIRACL languages including Hindi.
- **BGE-M3** [27] — selected for its native support of dense, sparse, and multi-vector retrieval within one model, allowing a secondary internal-consistency check against our separately implemented BM25 sparse signal.
- **LaBSE** [25] — included as a representative of embedding models optimized primarily for cross-lingual sentence similarity/bitext mining rather than retrieval specifically, to test whether retrieval-oriented pretraining (as in E5/BGE-M3) provides measurable benefit over similarity-oriented pretraining on this task.

The primary reported results use whichever of the three performs best on the development split; all three are reported in the retriever comparison table (Table III) for completeness.

E. Vector Database

FAISS [7] (specifically a flat or HNSW index, chosen based on the corpus-scale/latency trade-off measured in preliminary indexing trials) is used as the dense vector store, selected because it is free, runs fully locally without a hosted-service dependency, and is the most widely validated open-source approximate nearest-neighbor library in the retrieval literature.

F. Generation Pipeline

The generator is held fixed across all four retrieval strategies to isolate retrieval as the sole independent variable. We propose the **Gemini free-tier API** as the primary generator, with **Ollama-hosted open-weight models** (e.g., an open Hindi-capable instruction-tuned model run locally) evaluated as a secondary, fully offline-capable configuration to validate that conclusions are not an artifact of one specific generator. The prompt template is held identical across strategies: a fixed instruction in Hindi asking the model to answer strictly from the provided context, followed by the top- k retrieved chunks, followed by the query. k is held constant (e.g., $k = 5$) across all strategies within a given experimental run.

XI. Evaluation Methodology

Evaluation is organized around three metric families, summarized in **Table IV**.

Retrieval metrics: Precision@ k , Recall@ k , Mean Reciprocal Rank (MRR), normalized Discounted Cumulative Gain (nDCG@ k), Context Precision (Eq. 12), and Context Recall, computed against the IndicQA-derived relevance judgments where available and against LLM-judged relevance for the non-IndicQA portions of the corpus.

Generation metrics: Faithfulness (Eq. 13, RAGAS-style [33]), Answer Relevancy, Citation Accuracy (whether the answer’s claims can be traced to a specific retrieved chunk), Hallucination Rate (Eq. 14), ROUGE-L, BERTScore [35] (using a multilingual BERTScore variant appropriate for Hindi), and token-level F1 against IndicQA gold answers.

Efficiency metrics: end-to-end query latency (retrieval + generation, ms), peak memory usage during indexing and querying (GB), estimated per-query token cost (input + output tokens, relevant even under a free-tier API’s rate limits), and throughput (queries/second) under a single-worker consumer-hardware configuration.

All LLM-judged metrics (faithfulness, relevancy, context precision/recall, sufficiency) are computed using a judge-model prompting protocol consistent with the RAGAS framework [33], with the judge model configuration held identical across all four retrieval strategies to avoid introducing a judge-side confound.

XII. Baseline Comparison

Table II — Related Work Comparison

Study	Language(s)	Retrieval Strategies Compared	Hallucination Analysis	Adaptive Weighting	Hardware-Aware
Lewis et al., RAG [1]	English	Dense only	No	No	No
Karpukhin et al., DPR [6]	English	Dense vs. sparse (retrieval only, no generation)	No	No	No
Thakur et al., BEIR [10]	English	BM25, dense, hybrid (retrieval only)	No	No	No
MIRACL [19]	18 languages incl. Hindi	Sparse, dense (retrieval only)	No	No	No
Es et al., RAGAS [33]	Primarily English	Not retrieval-strategy focused	Yes (faithfulness)	No	No
This work (proposed)	Hindi	BM25, Dense, Hybrid, DAHR (full RAG pipeline)	Yes (isolated by retrieval strategy)	Yes (DAHR)	Yes

Table II is constructed from the properties reported in the cited papers and is intended to situate this work’s scope, not to claim superiority on metrics those papers did not attempt to report.

XIII. Experimental Results

Labeling notice. No experiments have yet been executed for this proposal-stage manuscript. All numeric values in this section are **illustrative and hypothetical**, constructed to communicate the *expected shape* of results consistent with the mechanisms described in Sections VII–VIII and with prior findings in the cited literature (e.g., hybrid retrieval outperforming single-strategy retrieval [8], [10]). They must not be cited or interpreted as observed experimental outcomes.

Table V — Illustrative Retrieval Performance (Hypothetical)

Strategy	Precision@5	Recall@5	MRR	nDCG@5
BM25	0.58	0.61	0.63	0.60
Dense (best embedding)	0.64	0.67	0.68	0.66
Static Hybrid ($\alpha = 0.7$)	0.69	0.72	0.73	0.71
DAHR (proposed)	0.74	0.76	0.78	0.75

Table VI — Illustrative Generation Performance (Hypothetical)

Strategy	Faithfulness	Answer Relevancy	ROUGE-L	BERTScore F1
BM25	0.71	0.68	0.42	0.81
Dense	0.75	0.73	0.45	0.83
Static Hybrid	0.79	0.77	0.48	0.85
DAHR (proposed)	0.84	0.81	0.51	0.87

Table VII — Illustrative Hallucination Rate Comparison (Hypothetical)

Strategy	Overall Hallucination Rate	Retrieval-Attributed	Generation-Attributed
BM25	0.29	0.19	0.10
Dense	0.25	0.14	0.11
Static Hybrid	0.21	0.10	0.11
DAHR (proposed)	0.16	0.06	0.10

Table VIII — Illustrative Latency and Memory Comparison (Hypothetical, single consumer GPU, e.g., 8GB VRAM)

Strategy	Mean Query Latency (ms)	Peak Query-time Memory (GB)	Indexing Time (relative)
BM25	45	0.6	1.0×
Dense	180	2.1	3.5×
Static Hybrid	210	2.3	3.6×
DAHR	225	2.4	3.6×

Table IX — Illustrative Query-wise Results by Category (Hypothetical nDCG@5)

Query Category	BM25	Dense	Static Hybrid	DAHR
Factoid	0.72	0.63	0.70	0.74
Definition	0.55	0.68	0.70	0.73
Comparative	0.41	0.66	0.68	0.72
Numerical	0.69	0.52	0.64	0.71
Multi-hop	0.38	0.61	0.63	0.69
Long-context	0.47	0.64	0.66	0.70

The pattern illustrated in Table IX is constructed to reflect the mechanism argued in Section VIII-A: BM25 is illustrated as strongest on lexically anchored categories (factoid, numerical) and weakest on paraphrase-dependent categories (comparative, multi-hop), Dense is illustrated with the inverse pattern, and DAHR is illustrated as matching or exceeding the better of the two in every category, which is the qualitative claim the proposed mechanism predicts and that actual experiments would need to confirm or refute.

XIV. Statistical Validation

Given the query-category-stratified design (Table IX), we propose the following validation protocol, to be applied once real experimental data is collected:

- **Confidence intervals.** Bootstrap resampling (1,000 re-samples) over the query set to construct 95% confidence intervals for each metric in Tables V–IX, per retrieval strategy.

- **Significance testing.** Paired testing across strategies, since all four strategies are evaluated on the identical query set and corpus: a paired t -test (or, given the non-normality typical of bounded metrics like nDCG and faithfulness, a Wilcoxon signed-rank test) between DAHR and each baseline, with Bonferroni correction applied across the three pairwise comparisons (DAHR vs. BM25, DAHR vs. Dense, DAHR vs. static Hybrid) to control family-wise error rate.
- **Variance analysis.** Reporting per-category variance in addition to means (Table IX), since a strategy with a high mean but high cross-category variance (e.g., Dense, strong on comparative/multi-hop but weak on factoid/numerical) represents a materially different deployment risk profile than a strategy with a lower mean but low variance.
- **Effect size.** Cohen’s d or rank-biserial correlation alongside p -values, since statistical significance alone is insufficient to argue practical importance at typical Hindi-QA evaluation set sizes.

No p -values or confidence intervals are reported in this manuscript, since doing so without underlying data would constitute exactly the kind of fabricated result this paper’s guidelines prohibit; Section XIII’s tables are illustrative only, and Section XIV specifies the validation protocol to be applied once genuine measurements exist.

XV. Error Analysis

We propose a dedicated error-analysis protocol classifying every hallucinated or incorrect answer into one of six categories, each associated with a specific expected failure mode per retrieval strategy:

1. **Retrieval Failure** — no chunk in the top- k contains the answer. Expected to be the dominant BM25 failure mode for comparative and multi-hop queries (Table IX), where the answer-bearing passage may not share surface vocabulary with the query.
2. **Wrong Context** — a topically related but non-answering chunk is retrieved with high confidence, misleading the generator. Expected to be a Dense-retrieval-characteristic failure, since embedding similarity can be high for topically adjacent but non-answering passages (a known dense-retrieval failure mode [6]).
3. **Missing Context** — the answer is split across multiple chunks and no single retrieved chunk contains it in full, even though some relevant material was retrieved. Expected to disproportionately affect multi-hop and long-context queries across all strategies, since it is primarily a chunking-granularity issue rather than a strategy issue (motivating the chunking ablation, Section XVI).
4. **Translation/Transliteration Error** — code-mixed Hindi-English query or corpus text (common in Indian government and educational documents) is mishandled by tokenization or embedding, producing a retrieval mismatch. Expected to be most visible in the public-document subset of the corpus (Table I).

5. **Ambiguous Query** — the query itself admits multiple valid interpretations, and retrieval optimizes for one interpretation not intended by the user. Expected to be roughly strategy-agnostic, i.e., a property of query authoring rather than retrieval mechanism.
6. **Generation Failure** — retrieval succeeded (sufficient context was present) but the generator nonetheless produced an unfaithful answer, e.g., by conflating entities or extrapolating beyond the provided context. This is the category isolated by the $\text{Suff} = 1$ branch of Eq. (14) and is expected to be roughly constant across retrieval strategies, since it is generator-attributable rather than retriever-attributable — a pattern loosely reflected in the “Generation-Attributed” column of Table VII remaining comparatively stable across strategies while “Retrieval-Attributed” hallucination declines from BM25 to DAHR.

Mitigations proposed for future work include: query rewriting/decomposition for multi-hop and ambiguous queries prior to retrieval; smaller, overlapping chunk sizes or semantic chunking to reduce Missing-Context errors; a dedicated Hindi-English code-mixed tokenizer or normalization step to reduce Translation/Transliteration errors; and generator-side self-consistency checking to reduce Generation Failure independent of retrieval strategy.

Table X — Error Analysis Summary (Expected Distribution, Illustrative)

Error Category	BM25 (%)	Dense (%)	Static Hybrid (%)	DAHR (%)
Retrieval Failure	34	18	14	9
Wrong Context	9	22	15	12
Missing Context	15	14	13	12
Translation/Transliteration	8	9	8	7
Ambiguous Query	11	11	11	11
Generation Failure	23	26	25	24

Percentages in Table X are illustrative proportions of total hallucinated/incorrect answers per strategy and are constructed to sum to 100% per column; they are not derived from executed experiments.

XVI. Ablation Study

To isolate the contribution of each DAHR component, we propose removing one component at a time from the full DAHR pipeline and re-measuring nDCG@5, Faithfulness, and Hallucination Rate:

- **(–) Adaptive Weight Prediction** — replace $\alpha(q)$ with the best fixed α from the static Hybrid baseline (reduces DAHR to static Hybrid; isolates the value of adaptivity itself, per the graceful-degradation property noted in Section VIII-C).
- **(–) Query Classification** — remove x_{type} from the feature vector (Eq. 5), refitting $\mathbf{w}_\alpha$ over the remaining five features, to isolate the marginal contribution of category prediction versus the lower-level lexical/syntactic/semantic features alone.

- (–) **Hybrid Retrieval (i.e., ablate fusion entirely)** — replace Eq. (9) with Dense-only or Sparse-only retrieval selected by a hard-thresholded $\alpha(q)$ (route to one retriever per query rather than blending both), testing whether soft fusion is necessary or whether hard routing suffices.
- (–) **Semantic Chunking** — where semantic chunking is used as an enhancement over fixed-size chunking (Section X-B), remove it and revert to pure fixed-size chunking, to quantify chunking’s contribution independent of retrieval strategy.
- (–) **Reranking** — where a cross-encoder reranking stage is applied after initial fusion, remove it, isolating the contribution of first-stage retrieval quality from second-stage reranking quality.

Table XIII — Ablation Study (Expected Contribution, Illustrative)

Configuration	nDCG@5	Faithfulness	Hallucination Rate
Full DAHR	0.75	0.84	0.16
– Adaptive Weight Prediction	0.71	0.79	0.21
– Query Classification	0.73	0.82	0.18
– Hybrid Fusion (hard routing)	0.70	0.80	0.20
– Semantic Chunking	0.72	0.81	0.19
– Reranking	0.73	0.82	0.18

Table XIII is constructed so that removing Adaptive Weight Prediction produces the largest illustrative degradation, consistent with this component being DAHR’s central claimed contribution; actual ablation experiments may not preserve this exact ordering, and reporting that possibility honestly is itself part of the intended contribution of running the ablation at all.

XVII. Discussion

The illustrative results in Sections XIII–XVI are constructed to be consistent with a specific, falsifiable hypothesis: that query-aware adaptive fusion improves retrieval and faithfulness precisely because different query categories have divergent lexical-versus-semantic matching requirements, and that a fixed global α necessarily under-serves whichever categories lie farthest from the tuned operating point. If borne out empirically, this hypothesis implies a broader methodological point for RAG system design: retrieval-strategy evaluation should be reported *per query category*, not only in aggregate, since aggregate metrics can mask large per-category regressions (as illustrated by BM25’s projected weakness on comparative/multi-hop queries in Table IX, which an aggregate-only report would obscure).

The accuracy–latency trade-off is expected to be real but modest in this design: DAHR’s added components (feature extraction, classification, weight prediction) are lightweight relative to the dense encoder forward pass and FAISS search that both static Hybrid and DAHR already require, so the illustrative latency gap between static Hybrid and DAHR in Table VIII (210ms $\rightarrow$ 225ms) is small relative to the gap between BM25 and any dense-involving strategy (45ms $\rightarrow$

180ms+). This suggests that, if confirmed, the primary latency cost in this design space is “using dense retrieval at all,” not “making the hybrid weight adaptive” — a materially useful distinction for deployment planning, since it implies DAHR’s benefits come at little additional cost once an organization has already committed to dense or hybrid retrieval.

The memory and cost profile favors BM25-only deployment where feasible: BM25 requires no embedding model, no GPU, and a smaller index footprint (Table VIII), making it the appropriate choice for extremely constrained deployments (e.g., low-end shared hosting with no GPU access) even though it is expected to underperform on paraphrase-dependent query categories. This is a genuine, unavoidable trade-off rather than one DAHR resolves, and Section XVIII (Limitations) and the deployment recommendations below treat it as such rather than overstating DAHR’s applicability to every hardware tier.

Deployment recommendation (conditional on confirmatory experiments). For organizations with at least a single consumer GPU (≥ 6 –8GB VRAM) and a query distribution known to include a meaningful share of comparative, definitional, or multi-hop questions — plausible for educational and public-scheme QA use cases — DAHR is the recommended strategy. For organizations restricted to CPU-only, no-GPU environments, BM25 remains the pragmatic default, with static Hybrid Retrieval as an intermediate option if a small, fixed embedding budget can be absorbed offline.

XVIII. Limitations

This work has the following limitations, stated candidly:

- **No experiments have yet been executed.** All numeric results in this manuscript are illustrative and hypothesis-consistent constructions, not empirical findings; the central claims (RQ1–RQ4) remain unverified pending implementation.
- **Single-language evaluation.** The proposed evaluation is Hindi-only; findings may not transfer to other Indian languages with different morphological complexity, script, or resource availability (e.g., lower-resource languages such as Santali or Bodo, which lack comparable pretrained embedding or QA resources).
- **Consumer-hardware scope.** Latency and memory findings are specific to the consumer-hardware tier described in Table VIII and may not generalize to multi-GPU production serving environments, where different bottlenecks (e.g., index sharding, concurrent-query throughput) dominate.
- **Reliance on free-tier APIs.** The Gemini free-tier API is subject to rate limits and possible model-version changes outside the authors’ control, which could affect exact reproducibility of generation-stage results over time; the Ollama-hosted secondary configuration partially mitigates this but at a cost in generation quality relative to larger hosted models.
- **Small-scale experimental design.** The proposed corpus (Table I) is modest relative to production-scale

Hindi corpora; findings at this scale may not extrapolate linearly to substantially larger indices.

- **Potential bias in LLM-judged metrics.** Faithfulness, relevancy, and sufficiency judgments (Eqs. 13–14) depend on an LLM judge that may itself have imperfect Hindi-language judgment quality or systematic biases, a limitation inherited from the RAGAS-style evaluation paradigm generally [33] and not specific to this framework.
- **Query classifier and weight predictor are trained on a specific query distribution.** DAHR’s adaptivity is only as good as the representativeness of the development set used to fit w_α (Eq. 15); a deployment query distribution that differs substantially from this development set (e.g., a different domain than the four corpus sources in Table I) may require recalibration.
- **Generalization across domains within Hindi itself is untested** by this design; the four corpus sources (Table I) were chosen for license availability and relevance, not because they are known to be representative of all Hindi QA use cases.

XIX. Future Work

Beyond executing the proposed experimental protocol itself, realistic extensions include: (1) evaluating DAHR on additional Indian languages with adequate embedding and QA resource availability (e.g., Bengali, Tamil, Marathi) to test cross-lingual generalization of the adaptive-weighting mechanism; (2) extending to explicitly cross-lingual retrieval settings where the query language and corpus language differ; (3) domain-specialized instantiations — Legal RAG for Indian case law and statutes, Healthcare RAG for public-health information dissemination, and Educational RAG for curriculum and examination support — each of which would require domain-specific query-category taxonomies distinct from the six general categories used here; (4) Agentic RAG extensions in which a query-planning agent issues multiple sub-queries and DAHR is applied per sub-query rather than once per top-level query; (5) GraphRAG-style structured retrieval over entity–relation graphs extracted from government and educational documents, potentially complementing rather than replacing DAHR’s chunk-level retrieval; (6) Multimodal RAG incorporating scanned government-document images and tables, common in real Indian public-sector corpora but entirely out of scope for the text-only design here; (7) learned, rather than linear, adaptive weight prediction (e.g., a small MLP or gradient-boosted model over the same six features), evaluated against the linear formulation of Eq. (8) to quantify the interpretability–performance trade-off explicitly; and (8) edge-deployment variants using quantized embedding models to further lower the hardware floor established in Table VIII.

XX. Conclusion

This paper has proposed Dynamic Adaptive Hybrid Retrieval (DAHR), a query-aware alternative to fixed-weight hybrid retrieval for Retrieval-Augmented Generation, motivated by

the well-established but under-exploited observation that different query categories place different demands on lexical versus semantic retrieval. We have provided a complete mathematical formulation of DAHR — query feature extraction, query classification, adaptive weight prediction, and a hybrid fusion objective linked explicitly to context precision, faithfulness, and hallucination probability — together with a fully specified, reproducible, open-source-only benchmarking protocol for comparing BM25, Dense, static Hybrid, and DAHR retrieval on Hindi question answering under consumer-hardware constraints. Because this manuscript is proposal-stage, its numeric results are explicitly illustrative rather than empirical; its contribution is the framework, the mathematical formulation, and the experimental design itself, which together are intended to close specific, identified gaps in Indian-language RAG evaluation (Section III) and to provide a concrete foundation for the implementation and subsequent empirical publication of this research direction.

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This manuscript is presented as a research proposal and implementation blueprint; Sections XIII, XV, and XVI contain illustrative, hypothesis-consistent values that are explicitly not the product of completed experiments, in accordance with the disclosure standard adopted throughout this paper.